

- 3 (a) temperature of the spheres is the same
no (net) transfer of energy between the spheres
- B1
B1 [2]
- (b) (i) power = $m \times c \times \Delta\theta$ where m is mass per second
3800 = $m \times 4.2 \times (42 - 18)$
 $m = 38 \text{ g s}^{-1}$
- C1
C1
A1 [3]
- (ii) some thermal energy is lost to the surroundings
so rate is an overestimate
- M1
A1 [2]

- 4 (a) straight line through origin
- M1